

EDUCATION

Massachusetts Institute of Technology, EECS PhD in Computer Science (Advised by Jacob Andreas)	Cambridge, MA <i>January 2025</i>
UNIVERSITY OF MICHIGAN ANN ARBOR, College of Engineering BSE in Computer Science with minor in Mathematics: Summa Cum Laude (GPA: 3.83)	Ann Arbor, MI <i>May 2025</i>

RESEARCH PUBLICATIONS

Pres, Ruis, Ghebreselassie, Li, Andreas (2026) Self-CTRL: Self-Consistency Training with Reinforcement Learning *Preprint*

- Introduced a method that optimizes for consistency between a model's self-explanations of behavior and its behavior
- Improved a third-party auditor's refusal-prediction accuracy from 36% to 92%

Pres*, Li*, Ruis*, et al.(2026) Position: It's Time to Optimize for Self-Consistency *The Forty-third International Conference on Machine Learning*

- Proposed self-consistency as a framework for understanding diverse LM failures, including sycophancy
- Showed that many existing techniques are special cases of a common consistency-optimization procedure

Stenseke, Kierans, **Pres**, Hadfield-Menell (2026) Position: Why LLMs Should Be Reasonably Morally Inconsistent *The Forty-third International Conference on Machine Learning - Workshop on Pluralistic Alignment*

- Argued that **moral consistency** should be a desirable but defeasible norm, with deviations made transparent

Lee, Bai, **Pres**, et al.(2024) A Mechanistic Understanding of Alignment Algorithms: A Case Study on DPO and Toxicity *The Forty-first International Conference on Machine Learning (Oral: top 1.5 % of submissions)*

- Discovered how toxicity is represented in pre-trained language models (GPT2, Llama2-7b)
- Studied the internal mechanisms of a popular alignment algorithm (Direct Preference Optimization)
- Discovered that DPO does not remove toxicity representations but rather suppresses them
- Uncovered why "aligned" models can be jailbroken

Pres, et al.(2024) Towards Reliable Evaluation of Behavior Steering Interventions in LLMs *The 38th Conference on Neural Information Processing Systems - Workshop on Foundation Model Interventions (Oral: top 5 submission)*

- Established four criteria for effective behavioral steering metrics and highlighted current metric shortcomings
- Developed new steering metric showing over-reported success of control method Contrastive Activation Steering

Park*, Lubana*, **Pres**, et al.(2024) Competition Dynamics Shape Algorithmic Phases of In-Context Learning *The 13th International Conference on Learning Representations (Spotlight)*

- Provided mechanistic analysis showing Transformers solve in-context learning with multiple competing algorithms
- Developed quantifiable techniques to measure neuron memorization of training data in transformers

PREVIOUS RESEARCH EXPERIENCE

Krueger AI Safety Lab Internship (University of Cambridge) <i>Mentored By David Krueger</i>	<i>May 2024 - Dec 2024</i>
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- Developed framework for multiple constraint satisfaction of Large Language Model generations at inference-time
- Explored generalizability of inference-time interventions such as Contrastive-Activation-Steering
- Developed novel benchmark to measure the success of behavioral steering interventions

Physics of Intelligence at the Center for Brain Science (Harvard University) <i>Mentored By Hidenori Tanaka, Ekdeep Singh Lubana</i>	<i>Sept 2024 - Present</i>
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- Rigorously studying Transformer model behaviors in thoughtfully constructed toy-settings
- Trained Transformers of different architectures to perform in-context learning (ICL)
- Researching the mechanisms these Transformers use to solve ICL tasks and handle underspecified queries
- Published findings at ICLR

University of Michigan Language and Information Technologies Lab <i>Supervised by Rada Mihalcea</i>	<i>October 2023 - Present</i>
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- Conducted research in mechanistic interpretability, with a focus on safety alignment algorithms
- Published research findings at ICML
- Researched Vision Language Model interpretability as well as In Context Learning mechanisms

*Mentored by **Neel Nanda***

- Analyzed and visualized high-dimensional LLM operations to generate hypothesis of LLM internal mechanisms
- Tested hypotheses of attention head superposition phenomenon in GPT2-XL
- Devised precise casual interventions to validate hypotheses under fact extraction contexts

Machine Learning Safety Scholars

June 2022 – August 2022

- Completed coding, written, and discussion assignments from MIT's Intro to Machine Learning, UM's Deep Learning for Computer Vision, and NYU's Deep Learning courses
- Distilled foundational machine learning papers into readable summaries
- Implemented safety techniques including adversarial and black swan robustness as well as interpretability

GRANTS AND AWARDS

National Science Foundation Graduate Research Fellowship (NSF GRFP)

April 2026

2024-25 University of Michigan Outstanding Undergraduate Research Award

May 2025

Effective Altruism Infrastructure Grant, Full stack Development

June 2022 – September 2022

- Developed prototype for web-app that increases transparency of unlikely global risk assessments
- Designed NoSQL MongoDB database structure for back-end
- Created interactive front-end using React.js
- Developed API connecting database to front-end using Node.js and Express.js